

2201NSC Linear Algebra and Applications

Week 9

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Applications of the SVD

Approximation, image compression and PCA

The SVD is an ordered sum of rank-one matrices

If $A \in \mathbb{R}^{m \times n}$ has rank r , then

$$A = U\Sigma V^T = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T, \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0.$$

Each outer product $\mathbf{u}_i \mathbf{v}_i^T$ is an $m \times n$ matrix of rank one.

Why the order matters

The singular value σ_i weights the i th contribution. Larger singular values identify the directions with the greatest influence on A .

Truncation gives a low-rank approximation

Keeping only the first k terms gives

$$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T, \quad k < r.$$

Equivalently, if U_k and V_k contain the first k singular vectors,

$$A_k = U_k \Sigma_k V_k^T = A V_k V_k^T.$$

What remains

The first k right-singular-vector directions and their contributions.

What disappears

All components in the directions $\mathbf{v}_{k+1}, \dots, \mathbf{v}_r$.

The singular values determine the approximation error

For a matrix $M = (m_{ij})$, the **Frobenius norm** is

$$\|M\|_F = \sqrt{\sum_{i,j} m_{ij}^2}.$$

It is the matrix analogue of Euclidean length and measures the combined size of all the entries. The SVD gives

$$\|A\|_F^2 = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2.$$

The error in the rank- k approximation is therefore

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \sigma_{k+2}^2 + \cdots + \sigma_r^2}.$$

As a proportion of the size of A , the relative error is

$$\frac{\|A - A_k\|_F}{\|A\|_F} = \sqrt{\frac{\sigma_{k+1}^2 + \cdots + \sigma_r^2}{\sigma_1^2 + \cdots + \sigma_r^2}}.$$

A rank- k SVD can reduce storage

An $m \times n$ matrix stores

mn numbers.

Storing U_k , the singular values, and V_k requires

$$mk + k + nk = k(m + n + 1) \text{ numbers.}$$

Storage decreases when

$$k(m + n + 1) < mn.$$

For an 800×1000 matrix, compression requires

$$k(800 + 1000 + 1) < 800000.$$

Song ratings as vectors

Eight users rate ten songs from 0 to 5. Place each user in one column of the rating matrix $R \in \mathbb{R}^{10 \times 8}$.

$$R = \begin{pmatrix} 0 & 5 & 3 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 5 & 3 & 2 \\ 5 & 4 & 4 & 5 & 4 & 0 & 3 & 3 \\ 5 & 5 & 3 & 3 & 4 & 4 & 3 & 2 \\ 0 & 0 & 0 & 0 & 3 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 4 & 5 & 0 & 2 \\ 0 & 0 & 5 & 2 & 3 & 3 & 0 & 1 \\ 5 & 3 & 3 & 2 & 4 & 2 & 1 & 1 \\ 0 & 0 & 0 & 5 & 1 & 3 & 1 & 0 \\ 0 & 4 & 4 & 3 & 4 & 1 & 0 & 3 \end{pmatrix}.$$

The new user has supplied

$$\mathbf{x} = (0, 0, 5, 0, 0, 0, 3, 3, 0, 0)^T.$$

Which existing users have the most similar preferences?

Normalisation compares preference patterns

Users who give generally high ratings should not automatically appear more similar.
Divide each user vector by its length:

$$\mathbf{q}_j = \frac{\mathbf{r}_j}{\|\mathbf{r}_j\|}, \quad \hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|}.$$

For the new user,

$$\|\mathbf{x}\| = \sqrt{5^2 + 3^2 + 3^2} = \sqrt{43},$$

so

$$\hat{\mathbf{x}} = \frac{1}{\sqrt{43}}(0, 0, 5, 0, 0, 0, 3, 3, 0, 0)^T.$$

Let $Q = (\mathbf{q}_1 \cdots \mathbf{q}_8)$.

Cosine similarity gives the original-space ranking

Because the vectors are normalised,

$$\cos \theta_j = \mathbf{q}_j^T \hat{\mathbf{x}}.$$

Stacking the eight dot products gives all similarities at once:

$$\begin{aligned} Q^T \hat{\mathbf{x}} &= (\mathbf{q}_1^T \hat{\mathbf{x}}, \dots, \mathbf{q}_8^T \hat{\mathbf{x}})^T \\ &\approx (0.704, 0.464, 0.732, 0.647, 0.625, 0.242, 0.405, 0.494)^T. \end{aligned}$$

Most to least similar

Users 3, 1, 4, 5, 8, 2, 7, 6.

This comparison gives greatest importance to matching individual songs.

The SVD identifies broader preference patterns

Transpose Q so that each observation occupies one row:

$$Q^T = U\Sigma V^T.$$

The columns of V are the principal directions in the ten-dimensional song space.

- ▶ $\mathbf{v}_1$ describes the strongest combined rating pattern.
- ▶ $\mathbf{v}_2$ describes the next strongest orthogonal pattern.
- ▶ The first two singular values are approximately 2.3216 and 0.9342.

Using only $\mathbf{v}_1$ and $\mathbf{v}_2$ gives a two-dimensional model of the dominant patterns in the ratings.

Multiplication gives the two principal-component coordinates

Let $V_2 = (\mathbf{v}_1 \ \mathbf{v}_2)$. The coordinate of User j is

$$\mathbf{q}_j^T V_2 = (\mathbf{q}_j^T \mathbf{v}_1, \mathbf{q}_j^T \mathbf{v}_2).$$

The rank-2 part of the SVD is

$$Q_2^T = U_2 \Sigma_2 V_2^T$$

The omitted right singular vectors are orthogonal to the columns of V_2 , so $Q^T V_2 = Q_2^T V_2$. Therefore

$$\begin{aligned} Q^T V_2 &= U_2 \Sigma_2 V_2^T V_2 \\ &= U_2 \Sigma_2 I_2 = U_2 \Sigma_2. \end{aligned}$$

The new user's coordinates are similarly calculated as $\hat{\mathbf{x}}^T V_2$.

The SVD places every user in a two-dimensional space

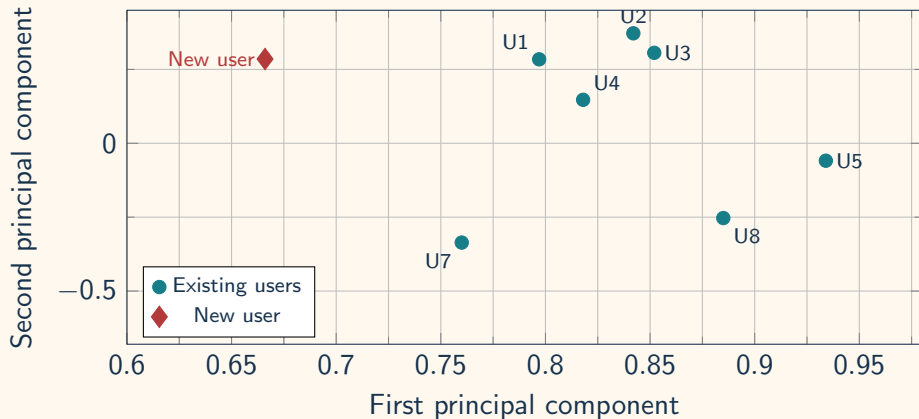
User	PC1	PC2	User	PC1	PC2
1	0.797	0.284	5	0.934	-0.059
2	0.842	0.372	6	0.645	-0.599
3	0.852	0.306	7	0.760	-0.336
4	0.818	0.147	8	0.885	-0.253

New user

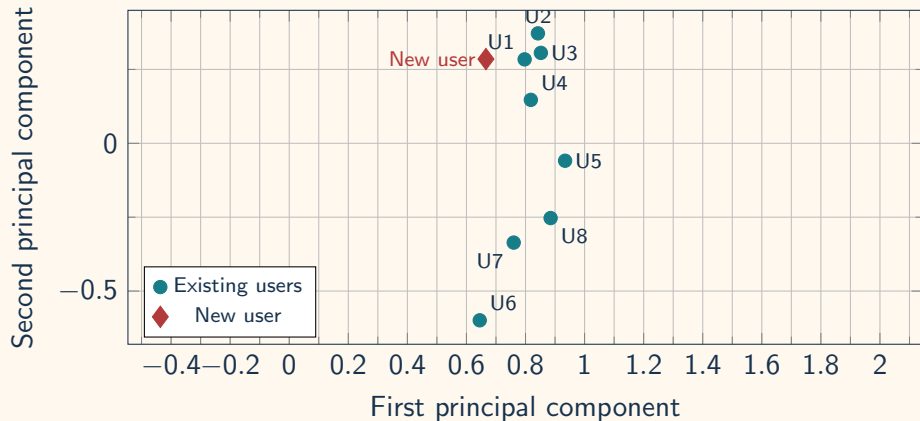
$$\hat{\mathbf{x}}^T V_2 = (0.666, 0.285).$$

The signs of either axis may reverse in another valid SVD. Distances and the resulting ranking do not change.

Independent axis scaling changes the visual impression



Equal axis scaling preserves Euclidean geometry



Distances in the reduced space give a new ranking

For User j at (a_j, b_j) and the new user at (a, b) ,

$$d_j = \sqrt{(a_j - a)^2 + (b_j - b)^2}.$$

For User 2,

$$d_2 = \sqrt{(0.842 - 0.666)^2 + (0.372 - 0.285)^2} \approx 0.196.$$

The distances to Users 1–8 are approximately

$$(0.131, 0.196, 0.187, 0.206, 0.436, 0.884, 0.627, 0.581).$$

Nearest to farthest

Users 1, 3, 2, 4, 5, 8, 7, 6.

The rankings differ because the SVD projection removes less important directions and Euclidean distance differs from cosine similarity.

Short Revision Quiz

Applications of the SVD

Question 1 — SVD expansion

Which expression is the rank-one contribution associated with σ_i ?

A. $\sigma_i \mathbf{u}_i \mathbf{v}_i^T$

B. $\mathbf{u}_i + \sigma_i + \mathbf{v}_i$

C. $\sigma_i UV^T$

Question 2 — Truncated SVD

Which expression gives the rank- k truncated-SVD approximation?

A. $A_k = \sum_{i=k+1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T$

B. $A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T$

C. $A_k = A - kI$

Question 3 — Approximation error

The approximation error for a rank- k approximation is given by

$$\|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \sigma_{k+2}^2 + \cdots + \sigma_r^2}.$$

If A has singular values 6, 3, and 1, what is $\|A - A_2\|_F$?

- A. 1
- B. 3
- C. $\sqrt{10}$

Question 4 — Storage

How many numbers are required to store a rank- k SVD approximation of an $m \times n$ matrix?

A. mnk

B. $m + n + k$

C. $k(m + n + 1)$

Question 5 — Normalising ratings

Why are the user-rating vectors normalised before their cosine similarities are calculated?

- A. To compare rating patterns without favouring users who give larger ratings
- B. To make every user give the same ratings
- C. To make the matrix square

Question 6 — Principal directions

Let $Q^T = U\Sigma V^T$. Which vectors define the new coordinate directions for the user-rating data?

- A. In the columns of U
- B. In the columns of V
- C. On the diagonal of Σ

Question 7 — Principal-component scores

Which matrix contains the first two coordinates of all eight existing users?

A. $Q^T V_2 = U_2 \Sigma_2$

B. $Q^T Q$

C. $V_2^T V_2 = I_2$

Question 8 — Arranging the observations

Why is Q transposed before applying PCA in the song-rating example?

- A. So that each row represents one user observation
- B. So that the matrix becomes symmetric
- C. So that the singular values become ratings

Question 9 — Projecting a new user

If $V_2 = (\mathbf{v}_1 \ \mathbf{v}_2)$, which expression gives the new user's two principal-component coordinates?

A. $V_2 \hat{\mathbf{x}}$

B. $\hat{\mathbf{x}}^T V_2$

C. $Q^T \hat{\mathbf{x}}$

Question 10 — Comparing rankings

Why can the reduced-space ranking differ from the cosine-similarity ranking?

- A. The SVD changes the users' original ratings at random
- B. Projection removes directions and the rankings use different distance measures
- C. The reduced space contains more dimensions than the original space

Question 11 — Comparing users in the reduced space

What does it suggest when two users are close together in the reduced principal-component space?

- A. They share similar dominant rating patterns
- B. They gave exactly the same rating to every song
- C. Their original rating vectors have the same length

Question 12 — Plot interpretation

When should visual distances in a score plot agree with Euclidean distances calculated from the coordinates?

- A. When both axes use equal unit lengths
- B. Whenever the plot fills the page
- C. Only when all scores are positive

Quiz answers 1–4

1 — A

The i th rank-one contribution is $\sigma_i \mathbf{u}_i \mathbf{v}_i^T$.

3 — A

Only $\sigma_3 = 1$ is omitted, so $\|A - A_2\|_F = 1$.

2 — B

The truncated SVD retains terms 1 through k .

4 — C

U_k , the singular values, and V_k store $mk + k + nk$ numbers.

Quiz answers 5–8

5 — A

Normalisation compares the direction of each rating vector rather than its overall size.

6 — B

The right singular vectors, which are the columns of V , are the principal directions for the song-rating coordinates.

7 — A

Multiplying Q^T by V_2 gives the first two coordinates of all eight users: $Q^T V_2 = U_2 \Sigma_2$.

8 — A

After transposing Q , each row of Q^T represents one user.

Quiz answers 9–12

9 — B

Post-multiplying the new user's row vector by V_2 gives its two coordinates.

10 — B

The projection discards directions outside the first two principal vectors, and Euclidean distance differs from cosine similarity.

11 — A

Nearby users have similar coordinates in the dominant rating patterns retained by the SVD. Their individual song ratings need not be identical.

12 — A

Equal unit lengths preserve the geometry of Euclidean distance in the plot.

Final takeaway

The SVD orders matrix information by singular value. Truncation supports approximation, while PCA uses the leading directions to describe and compare data.